Highlights

Topology Optimization of Heat Exchangers Using Physics-Informed Neural Networks

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- PINN-TO addresses the computational efficiency challenges of traditional TO methods.
- The framework enables efficient optimization of heat exchanger geometries.
- Experimental validation demonstrates the effectiveness of the PINN-TO approach.

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John A. Gardiner[a,*], Hailei Wang[a]

[a]Department of Mechanical Engineering, Utah State University, Old Main Hill, Logan, 84322, UT, USA

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ABSTRACT

Topology optimization (TO) of heat exchangers is a promising approach for enhancing thermal-hydraulic performance in various engineering applications. However, traditional TO methods often rely on computationally expensive physics solvers, limiting their practicality for complex designs and widespread application. This research proposes the integration of Physics-Informed Neural Networks (PINNs) with TO to create a novel framework for heat exchanger optimization. The PINN-TO framework will leverage the ability of PINNs to quickly approximate solutions to governing equations of fluid flow and heat transfer, enabling efficient evaluation and optimization of heat exchanger geometries. The objectives of this research include developing the PINN-TO framework, quantifying its performance against traditional TO methods, validating the optimized designs through experimental testing, and generalizing the framework across various heat exchanger configurations. The anticipated outcome is a computationally efficient and effective method for designing high-performance heat exchangers, contributing to advancements in customized heat transfer applications.

1. Introduction

P1: What is heat exchanger topology optimization? How does it improve the performance of heat exchanger designs? What are the challenges and limitations of traditional topology optimization methods in the context of heat exchanger design? Talk about computational expense and the difficulty of manufacturing the optimized designs.

P2: How can Physics-Informed Neural Networks (PINNs) be used to address the challenges of traditional topology optimization methods in heat exchanger design? What are the advantages of using PINNs in this context? How do PINNs work in general and what makes them suitable for this application? Introduce the concept of surrogate models and their role in accelerating the optimization process.

P3: Where can this PINN-TO approach be applied in the field of heat exchanger design? What are the potential applications and benefits of using this method in real-world scenarios?

2. Methodology

This is where I will describe the methodology and structure of the PINN TO model.

2.1. PINN-TO Structure

Introduce the structure of the PINN-TO system. Talk about the geometry generator model, the PINN simulator model, and the optimization algorithm. Talk about the two stage generator training process including the reinforcement learning aspect.

2.2. Project Specifics

Describe the details of the testing setup. Explain the domain size, the boundary conditions, fluid properties, and the simulation parameters.

2.3. Data Collection

Describe the process of collecting the data needed for the study. Explain the sources of the data and the methods used to gather and preprocess it.

2.4. Reinforcement Learning and Generator Training

Describe the reinforcement learning approach used to train the generator model. Explain how the generator is trained to produce optimal geometries and how the reinforcement learning algorithm is integrated with the overall PINN-TO framework.

3. Results

This is where I will show the results of the base case and demonstrate the effectiveness of the PINN TO approach.

3.1. Base Case

Show the results of the base case, including any relevant metrics or visualizations.

3.2. PINN accuracy vs training dataset size

Show the relationship between the accuracy of the PINN model and the size of the training dataset.

3.3. Geometry grid resolution effect

How do the geometry grid resolution and the resulting accuracy of the PINN model compare?

*Corresponding author
austin.gardiner@usu.edu (J.A. Gardiner); hailei.wang@usu.edu (H. Wang)
ORCID(s): 0000-0003-4965-8076 (J.A. Gardiner)

3.4. More representative test case

Show the results of a more representative test case, demonstrating the effectiveness of the PINN-TO approach in a more realistic scenario.

4. Conclusion

This is where I will summarize the findings and discuss the implications of the research.

CRediT authorship contribution statement

John A. Gardiner: Investigation, Methodology, Validation, Writing - original draft. **Hailei Wang:** Supervision, Writing - review and editing.

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